YAKEEN

NEET 2.0 2027

Physics | Lecture - 7

Motion in a Straight Line

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Todays Goal

Questions Practice

Vali Concept

$\int a\,dt$ = change in velocity

$\int v\,dt$ = displacement

Area of x/t graph

Position $\vec{x}$

$\vec{v}$

$\vec{a}$

$\int a\,dt$

$\int v\,dt$

diff^n

$$\vec{v} = \frac{d\vec{x}}{dt}$$

velocity = Slope of x/t graph

diff^n of v(t)

$$\vec{a} = \frac{d\vec{v}}{dt}$$

diff^n = Slope of v/t is $\left(\frac{dv}{dt}\right)$

gf v(x)

$$\vec{a} = v\frac{dv}{dx}$$

$\int$ of $\int a\,dx$

$$\frac{d\vec{a}}{dt} = \text{Jerk}$$

Not in Syllabus

Slope of a/t graph = $\int$ of $\int a\,dx$

$$\Delta x = \int_{x_i}^{x_f} dx = \int v\,dt = \text{Area of } v/t \text{ graph}$$

$$\Delta v = \int_{u}^{f} dv = \int_{t_i}^{t_f} a\,dt = \text{Area of } a/t \text{ graph}$$

gm Which case acceleration is uniform (constant)

(1) $x = \sqrt{t} + 4$ ✗

(2) $\sqrt{x} = t - 3$ ✓

(3) $x = t^{-2} + 4$ ✗

(4) $x = \frac{4}{t}$ ✗

(5) $x = t^3 + 4$ ✗

(6) $x = \frac{3}{t^2}$ ✗

(7) $x = 4t^2 + 6$ ✓

(8) $V = 3t^2 + 4$ ✗

(9) $V = 6t - 4$ ✓

(10) $V = \sqrt{x}$ ✓

(11) $V = x^2 + 4$ ✗

(12) $V = 2x + 3$ ✗

(13) $a = 2t$ ✗

(14) $a = 3x$ ✗

(15) $a = 2t^2$ ✗

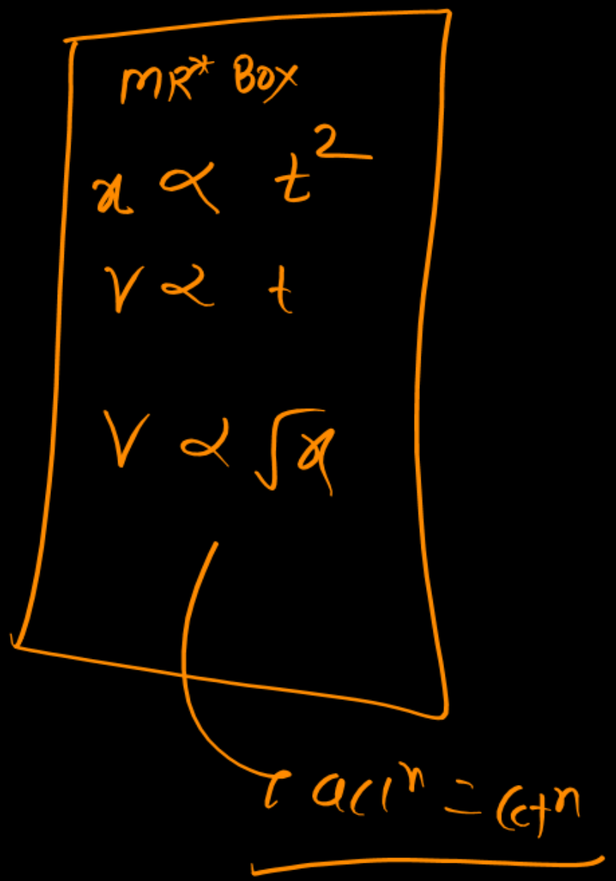
(16) $a = 5 m/s^2$ ✓

(17) $a = 3\sqrt{x}$ ✗

(18) $x = A\sin(\omega t)$ ✗

(19) $V = A\cos(\omega t)$ ✗

(20) $a = 2\sin(t)$ ✓ ✗


MR* Box
$x \propto t^2$
$v \propto t$
$v \propto \sqrt{x}$
Tacit ^{n} = Ctn

Question

If velocity $V = k\sqrt{x}$ then which of the following is correct for position [IIT-2006]

1. $x \propto \sqrt{t}$
2. $x \propto t^3$
3. $x \propto t^{-2}$
4. $x \propto t^2$

Question

If the displacement of a particle varies with time as $\sqrt{x} = t + 7$, then

1. Velocity of the particle is inversely proportional t
2. Velocity of the particle is proportional to t^2
3. Velocity of the particle is proportional to $\sqrt{t}$
4. The particle moves with constant acceleration

Velocity of object $v = 2x^3$ then find acceleration ??

Copy

$$a = v\frac{dv}{dx}$$

$$= 2x^3 \frac{d(2x^3)}{dx}$$

$$= 2x^3 \times 2 \times 3x^2$$

$$= 12x^5$$

Question

If position $x = t^2 + 5t^3 + 6$ then find

(i) Initial acceleration. $(t=0)$

(ii) Initial velocity. $(t=0)$

(iii) Acceleration at $t = 2$ sec.

$x = t^2 + 5t^3 + 6$

$\rightarrow V = \frac{d(t^2+5t^3+6)}{dt}$

$= 2t + 5\times3t^2 + 0$

$V = 2t + 15t^2$ ✓

$V(t=0) = 2\times0 + 15\times0$

$= 0\ m/s$

$a = 2 + 15(2t) = 2 + 30t$

$a(t=0) = 2\ m/s^2$

$\therefore a(t=2) = 2 + 30\times2 = 62\ m/s^2$

Question

If position $x = at^2 - bt^3$ find the acceleration is zero.

$x = at^2 - bt^3$

$\frac{dx}{dt} = v = a(2t) - b\,3t^2$

$a = \frac{dv}{dt} = \frac{d(2at - 3bt^2)}{dt}$

$= 2a\frac{dt}{dt} - 3b\frac{dt^2}{dt}$

$= 2a - 3b(2t)$

$a = 2a - 6bt$

Put $a = 0$

$0 = 2a - 6bt$

$2a = 6bt$

$t = \frac{2a}{6b} = \frac{a}{3b}$

(Q) Position of object $x = t^3 - 6t^2 + 10$ Find time when acceleration of object will be zero.

Copy

(Position) $x = t^3 - 6t^2 + 10$

$V = 3t^2 - 6(2t) + 0$

PW a=0

$a = \frac{dv}{dt} = 3(2t) - 12$

$0 = 6t - 12$

$12 = 6t$ $t = 2$ sec

Position of object $x = 2t^3 - 4t^2 + 4$

No ans find velocity, acceleration and Position at $t = 2$ Sec ✓

Soln $x = 2t^3 - 4t^2 + 4$

$$V = \frac{d(x)}{dt} = \frac{d(2t^3 - 4t^2 + 4)}{dt}$$

$$= 2(3t^2) - 4(2t) + 0$$

$$V = 6t^2 - 8t$$

$$a = 6(2t) - 8$$

$a = 12t - 8$

$V = 6t^2 - 8t$

$x = 2t^3 - 4t^2 + 4$

$t = 2$ sec ✓

Put

Position $x = t^3 - 9t^2 + 4$. find velocity?? when acceleration is zero;

copy

find $V = ??$ at a time when $a = 0$

Soln

$x = t^3 - 9t^2 + 4$

$V = 3(t^2) - 9(2t) + 0$

$V = 3t^2 - 18t$

$a = \frac{dv}{dt} = 3(2t) - 18 = 0$

$6t - 18 = 0$

$t = \frac{18}{6} = 3\,sec$

$V = 3(3)^2 - 18 \times 3 = 27 - 54 = -27\,m/s$

Position of object $x = 2t^3 - 24t + 4$ find acceleration when velocity of object is zero.

Soln

find $a = ??$ at time when $v = 0$

$x = 2t^3 - 24t + 4$

$v = 2(3t^2) - 24$

$V = 6t^2 - 24$ —— (*)

Put $V = 0 = 6t^2 - 24$

$24 = 6t^2$

$t^2 = 4$

$t = \sqrt{4}$

$t = 2\,sec$ ✓

$a = \frac{dv}{dt} = 6(2t) - 0 = 12t$

$a = 12 \times 2 = 24\,m/s^2$ ✓

Question

The position x of particle moving along x-axis varies with time t as $x = A \sin(\omega t)$ where A and w are 1 & 2 positive constants. The acceleration a of particle varies with its position (x) as

1. $a = Ax$
2. $a = -\omega^2 x$
3. $a = A\,\omega\,x$
4. $a = \omega^2 x\,A$

cusp

$X = A\sin(\omega t)$

$V = \frac{dx}{dt} = \frac{d A\sin(\omega t)}{dt}$

$= A\frac{d\sin(\omega t)}{dt}$

$= A\cos(\omega t) \times \omega$

$V = A\omega\cos(\omega t)$

$a = \frac{dv}{dt} = A\omega\left[-\sin(\omega t)\times\omega\right]$

$a = -A\omega^2\sin(\omega t)$

$a = -\omega^2 x$

SHM

$a(x) = ?$

Question

Copy

The initial velocity of a particle is u (at $t = 0$) and the acceleration a is given by $\alpha t^{3/2}$. Which of the following relations is valid?

1. $v = u + \alpha t^{3/2}$
2. $v = u + \frac{3\alpha t^3}{2}$
3. $v = u + \frac{2}{5}\alpha t^{5/2}$ (+1)
4. $v = u + \alpha t^{5/2}$ (-1)

$t=0 \rightarrow u$ ✓

Imp

$a = \alpha t^{3/2}$ (variable)

$\frac{dv}{dt} = \alpha t^{3/2}$

$\int_u^v dv = \int_{t=0}^{t} \alpha t^{3/2}\, dt$

$[v]_u^v = \alpha \frac{t^{\frac{3}{2}+1}}{\frac{3}{2}+1}$

$v = u + at$

$v = u + \alpha t^{3/2} \cdot t^1$

$v = u + \alpha t^{5/2}$

MR Scam (-1)

$v - u = \frac{\alpha t^{5/2}}{5/2}$

$v = u + \frac{2}{5}\alpha t^{5/2}$ ✓

Question

A particle moves along a straight line such that its displacement at any time t is given by $s = (t^3 - 6t^2 - 3t + 4)$ meters. The velocity when the acceleration is zero is

1. 3 m/s
2. 42 m/s
3. – 9 m/s
4. – 15 m/s

$s = t^3 - 6t^2 + 3t + 4$

$V = 3t^2 - 12t + 3$

$= 3(4) - 12 \times 2 + 3$

$= 12 - 24 + 3$

$= -12 + 3 = -9$

$a = 6t - 12 = 0$

$t = 2$

done

Question

NO

Position of object $x = t^3 - 6t^2 + 10$. Find time when acceleration of object will be zero.

$X = t^3 - 6t^2 + 10$

$a = 0$

done

Question

Position of object $x = 2t^3 - 4t^2 + 4$. Then, find velocity, acceleration and position at $t = 2$ sec?

Question

NO

The distance travelled by a particle is related to time t as $x = 4t^2$. The velocity of the particle at $t = 5s$ is: [25 Jan, 2023]

1. $40\ ms^{-1}$
2. $25\ ms^{-1}$
3. $20\ ms^{-1}$
4. $8\ ms^{-1}$

$x = 4t^2$

$v = \frac{dx}{dt} = \frac{d(4t^2)}{dt}$

$= 4(2t)$

$= 8t$

Question

The position of a particle related to time is given by $x = (5t^2 - 4t + 5)$m. The magnitude of velocity of the particle at $t = 2$s will be: **[15 April, 2023]**

1. $10\ ms^{-1}$
2. $14\ ms^{-1}$
3. $16\ ms^{-1}$
4. $06\ ms^{-1}$

$X = 5t^2 - 4t + 5$.

No

Question

The distance travelled by an object in time t is given by $s = (2.5)t^2$. The instantaneous speed of the object at $t = 5$s will be: **[13 April, 2023]**

1. $12.5\ ms^{-1}$
2. $62.5\ ms^{-1}$
3. $5\ ms^{-1}$
4. $25\ ms^{-1}$

$S = 2.5\,t^2$ ✓

$S = 2.5\,(2t)$

$= 5t$

N

Question

Copy

The velocity of a particle is $v = v_0 + gt + Ft^2$. Its position is $x = 0$ at $t = 0$; then its displacement after time ($t = 1$) is: **[17 March, 2021]**

1. $v_0 + g + f$
2. $v_0 + \frac{g}{2} + \frac{F}{3}$
3. $v_0 + 2g + 3F$
4. $v_0 + \frac{g}{2} + F$

velocity

$V = V_0 + gt + ft^2$

$\frac{dx}{dt} = V_0 + gt + ft^2$

$\int dx = \int_0^1 V_0\,dt + \int gt\,dt + \int ft^2\,dt$

$\Delta x = V_0 + \frac{g_0}{2} + \frac{f}{3}$

$\Delta x = (V_0 t)_0^1 + \left(g\frac{t^2}{2}\right)_0^1 + \left(f\frac{t^3}{3}\right)_0^1$

Question

The position of a particle as a function of time ts, is given by $x(t) = at + bt^2 - ct^3$ where a, b and c are constants. When the particle attains zero acceleration, then its velocity will be:

[09 April, 2019]

1. $a + \frac{b^2}{4c}$
2. $a + \frac{b^2}{c}$
3. $a + \frac{b^2}{2c}$
4. $a + \frac{b^2}{3c}$

$X = at + bt^2 - ct^3$ ✓

done

Question

The displacement 'x' (in meter) of a particle of mass 'm' (in kg) moving in one dimension under the action of a force, is related to time 't' (in sec) by $t = \sqrt{x} + 3$. The displacement of the particle when its velocity is zero, will be **[NEET 2013]**

1. 4 m
2. 0 m (zero)
3. 6 m
4. 2 m

$t = \sqrt{x} + 3$

$\sqrt{x} = t - 3$

$x = (t-3)^2 = t^2 - 2\times3t + 9$

$x = t^2 - 6t + 9$

$V = 2t - 6 + 0$

$V = 0 = 2t - 6 \quad (t = 3)$

$x = (t-3)^2$

$x_{(t=3)} = (3-3)^2 = 0$

Question

A particle moving along x-axis has acceleration f, at time t, given $f = f_0\left(1 - \frac{t}{T}\right)$, Where f_0 and T are constants. The particle at $t = 0$ has zero velocity. At the instant when $t = T$ the particle's velocity is **[AIPMT (Prelims)-2007]**

$t=0 \rightarrow u=0$

1. $\frac{1}{2} f_0 T$
2. $f_0 T$
3. $\frac{1}{2} f_0 T^2$
4. $f_0 T^2$

$$V = f_0 (t)_0^T - \frac{f_0}{T}\left(\frac{t^2}{2}\right)_0^T$$

$$= f_0 T - \frac{f_0}{2T}(T^2 - 0)$$

$$= f_0 T - \frac{f_0 T}{2}$$

$$= \frac{f_0 T}{2}$$

$$\text{acc}^n = f_0\left(1 - \frac{t}{T}\right)$$

$$\frac{dv}{dt} = \left(f_0 - \frac{f_0}{T} t\right)$$

$$\int_0^v dv = \int_0^T f_0\, dt - \int_0^T \frac{f_0}{T} t\, dt$$

Question

A body is moving with variable acceleration (a) along a straight line. The average acceleration of body in time interval t_1 to t_2 is

1. $\frac{a[t_2 + t_1]}{2}$ ✗
2. $\frac{a[t_2 - t_1]}{2}$ ✗
3. $\frac{\int_{t_1}^{t_2} a\,dt}{t_2 + t_1}$ ✓ Ans
4. $\frac{\int_{t_1}^{t_2} a\,dt}{t_2 - t_1}$

$$\langle a \rangle = \frac{\int_{t_1}^{t_2} a\,dt}{\int_{t_1}^{t_2} dt} = \frac{\int_{t_1}^{t_2} a\,dt}{t_2 - t_1}$$

Question

Copy

A particle moves in a straight line and its position x at time t is given by $x^2 = 2 + t$. Its acceleration is given by

1. $\frac{-2}{x^3}$
2. $-\frac{1}{4x^3}$
3. $-\frac{1}{4x^2}$
4. $\frac{1}{x^2}$

$x^2 = 2 + t$

diff^n w.r.t. x both side

$\frac{dx^2}{dx} = \frac{d2}{dx} + \frac{dt}{dx}$

$2x = 0 + \frac{1}{v}$

$2x = \frac{1}{v}$

$v = \frac{1}{2x}$

$\frac{dx^2}{dt} = \frac{d2}{dt} + \frac{dt}{dt}$

$2x\left(\frac{dx}{dt}\right) = 0 + 1$

$2xv = 1$

$v = \frac{1}{2x}$

$a = \frac{dv}{dt}$ ✗

$a = v\frac{dv}{dx}$

$a = \frac{1}{2x}\frac{d\left(\frac{1}{2}x^{-1}\right)}{dx}$

$= \frac{1}{4x}\left(-1x^{-1-1}\right)$

$= \frac{-1}{4x\,x^2} = \frac{-1}{4x^3}$ ✓

Question

Gp

A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$. The acceleration of particle is proportional to [2010]

1. (velocity)$^{3/2}$
2. (distance)^2 ✗
3. (distance)$^{-2}$ ✗
4. (velocity)$^{2/3}$ ✗

$a \propto (t+5)^{\frac{-3\times2}{2}}$

$a \propto (t+5)^{\frac{-2\times3}{2}}$

$a \propto V^{3/2}$

$x = (t+5)^{-1}$

outside t

$\frac{dx}{dt} = v = -1(t+5)^{-2} \times (1+0)$

$V = -1(t+5)^{-2}$

$a = \frac{dv}{dt} = -1(-2)(t+5)^{-3} \times 1$

$a = +2(t+5)^{-3}$

$a \propto (x)^3$

Question

If acceleration of object $a = 2x^{3/2}$ then find velocity at x where initial at $x = 0$ is 4 m/s

velocity

$a = 2x^{3/2}$ (varible)

$\frac{dv}{dt} = 2x^{3/2}$

$\int dv = \int 2x^{3/2}\,dt$

wrong

$v\frac{dv}{dx} = 2x^{3/2}$

$\int_4^v v\,dv = \int_{x=0}^{x} 2x^{3/2}\,dx$

$\left(\frac{v^2}{2}\right)_4^v = 2\left(\frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1}\right)_0^x = 2\,\frac{x^{5/2}}{5/2} = \frac{4}{5}x^{5/2}$

$\frac{v^2-16}{2} = \frac{4}{5}x^{5/2}$

$v^2 = 16 + \frac{8}{5}x^{5/2}$

$v = \left(16 + \frac{8}{5}x^{5/2}\right)^{1/2}$

Question

Copy

IIT-Adv-2013

The relation between time t and distance x is $t = \alpha x^2 + \beta x$ where α and β are constants. The retardation is:

1. $2\alpha v^3$
2. $2\beta v^2$
3. $2\alpha\beta v^2$
4. $2\beta^3 v^3$

$a = \frac{dv}{dt}$ ✗

$a = v\frac{dv}{dx}$

$= \frac{1}{2\alpha x+\beta} \times \frac{d(2\alpha x+\beta)^{-1}}{dx}$

$= \frac{-1(2\alpha x+\beta)^{-2} \times 2\alpha}{(2\alpha x+\beta)}$

$= \frac{-2\alpha}{(2\alpha x+\beta)^3} = -2\alpha\left(\frac{1}{2\alpha x+\beta}\right)^3 = -2\alpha v^3$

$t = \alpha x^2 + \beta x$

diff^n w.r.t x

$\frac{dt}{dx} = \alpha 2x + \beta \times 1$

$\frac{1}{v} = 2\alpha x + \beta$

$V = \frac{1}{2\alpha x+\beta} = (2\alpha x+\beta)^{-1}$

direction

$t = \alpha x^2 = \beta x$ ✓

$a = 2\alpha v^3$

$= \frac{t}{x^2}\left(\frac{x}{t}\right)^3$

$= \frac{t}{x^2} \times \frac{x^3}{t^3} = \frac{x}{t^2}$

Question NO

If $a = 3t^2 + 2t$, initial velocity is 5 m/s. Find the velocity at $t = 4$s. The motion is in straight line, a is acceleration in m/s^2 and t is time in seconds.

$t=0$
$5\ m/s$

$a = 3t^2+2t$ $\quad$ $v(t=4)$

(time varying)

$a = \frac{dv}{dt}$

$\int dv = \int a\,dt$

$\int dv = \int (3t^2+2t)\,dt$

$\frac{dv}{dt} = 3t^2+2t$

$\int_5^v dv = \int_{t=0}^{4} (3t^2+2t)\,dt$

$v-5 = \left(\frac{3t^3}{3}\right)_0^4 + \left(\frac{2t^2}{2}\right)_0^4$

$v-5 = 4^3 + (4)^2$

$= 64 + 16$

$v-5 = 80$

$v = 85\ m/s$

Question

Cell

JEE-Adv

The deceleration experienced by a moving motorboat after its engine is shut-off is given by $dv/dt = -kv^3$, where k is a constant. If v_0 is the magnitude of the velocity at shut-off, find the velocity as a function of t.

$$\int \frac{1}{x^3}\,dx = \int x^{-3}\,dx = \frac{x^{-3+1}}{-3+1}$$

$t = 0 \to V_0$

$V(t) = ??$

$$\frac{dv}{dt} = -kv^3$$

$$\int_{V_0}^{V} \frac{dv}{v^3} = -\int_0^t k\,dt$$

$$\left(\frac{v^{-3+1}}{-3+1}\right)_{V_0}^{V} = -(kt)_0^t$$

$$\left(\frac{v^{-2}}{-2}\right)_{V_0}^{V} = -kt$$

$$\left(\frac{1}{v^2}\right)_{V_0}^{V} = 2kt$$

$$\frac{1}{v^2} - \frac{1}{V_0^2} = 2kt$$

$$\frac{1}{v^2} = 2kt + \frac{1}{V_0^2}$$

$$\frac{1}{v^2} = \frac{2ktV_0^2 + 1}{V_0^2}$$

$$v^2 = \frac{V_0^2}{2ktV_0^2 + 1}$$

$$v = \frac{V_0}{\sqrt{2ktV_0^2 + 1}}$$

Question

Calculus JEE-Adv

The motion of a body is given by $dv/dt = 6 - 3v$, where v is in m/s. Find

(a) the velocity in terms of t and

(b) terminal velocity. The motion starts from rest.

$v = 2$ m/s

Method

$\frac{dv}{dt} = 6 - 3v = a$

$v_T = ?$ ($\frac{dv}{dt} = 0$)

$0 = 6 - 3v$

$3v = 6$

$v = 2$ m/s

Given

$\frac{dv}{dt} = 6 - 3v$

$\int_0^v \frac{dv}{(6-3v)} = \int_0^t dt$

$\left[\frac{\log(6-3v)}{-3}\right]_0^v = t$

$\left[\log(6-3v)\right]_0^v = -3t$

$\log(6-3v) - \log(6) = -3t$

$\log_e\left(\frac{6-3v}{6}\right) = -3t$

Anti-log

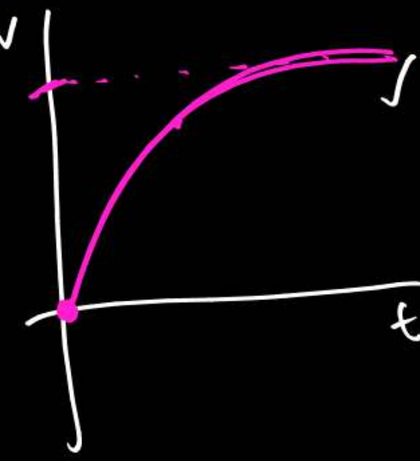
$e^{-3t} = \frac{6-3v}{6}$

$6e^{-3t} = 6 - 3v$

$3v = 6 - 6e^{-3t}$

$3v = 6(1 - e^{-3t})$

$v = 2(1 - e^{-3t})$



NEET - 2024

x (Position) $= at^4 + \beta t^2 + \gamma t + \delta$ find ratio of intial velocity to intial acceleration

(t=0) (t=0)

*

$$x = at^4 + \beta t^2 + \gamma t + \delta$$

$$\frac{dx}{dt} = V = a\,4t^3 + \beta(2t) + \gamma$$

$$V = 4at^3 + 2\beta t + \gamma$$

$$a = 4a(3t^2) + 2\beta \times 1$$

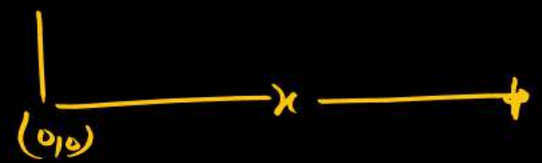
$$V_{t=0} = \gamma$$

$$a_{t=0} = 2\beta$$

$$\frac{V_{t=0}}{a_{t=0}} = \frac{\gamma}{2\beta}$$

(8) NEET - 2006

distance of Particle from origin given by $x = 40 + 12t - t^3$ How long Particle moves before coming to rest. ($v=0$)



$x = 40 + 12t - t^3$ ✓

$V = 0 + 12 - 3t^2 = 0$

$y^2 = x + 2$

$t = \sqrt{4} = 2s$

$x_{t=2} = 40 + 12\times2 - (2)^3$

$= 40 + 24 - 8$

$= 56$

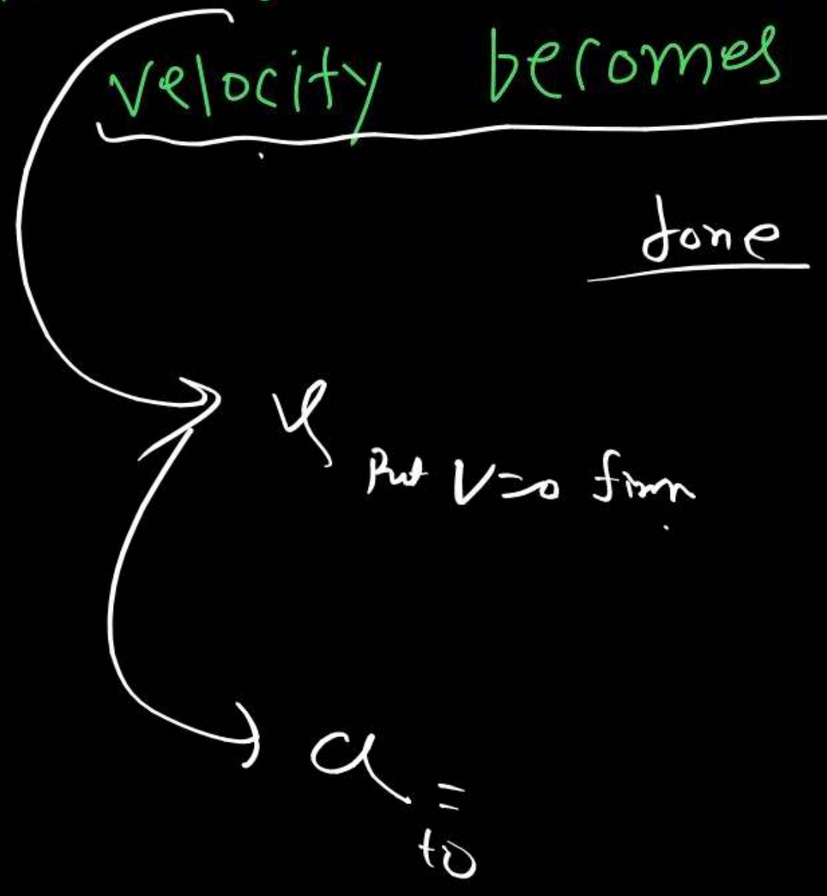
$x_{t=0} = 40$

$x_f - x_i = 56 - 40 = 16m$ ✓

NEET - 2012

Position $x = 8 + 12t - t^3$ find retardation of Particle when velocity becomes zero.

done



NEET - 2016

Velocity $V = At + Bt^2$ where A and B are constant then distance travelled between 1s to 2sec.

Soln

$V = At + Bt^2$

$a = \frac{dv}{dt}$

$a = A1 + B(2t)$

$\frac{dx}{dt} = At + Bt^2$

$\int dx = \int_1^2 At\,dt + \int_1^2 Bt^2\,dt$

$x = A\left(\frac{t^2}{2}\right)_1^2 + B\left(\frac{t^3}{3}\right)_1^2$

$= \frac{A}{2}\left(2^2 - 1^2\right) + \frac{B}{3}\left(2^3 - 1^3\right)$

$= \frac{3A}{2} + \frac{7B}{3}$

NEET - 2015

velocity $V = \beta x^{-2n}$ then find acceleration as a function of 'x'.

$$V = \beta x^{-2n}$$

$$a = V \frac{dv}{dx}$$

$$= \beta x^{-2n} \frac{d\, \beta x^{-2n}}{dx}$$

$$= \beta^2 x^{-2n} (-2n) x^{-2n-1} = -2n\beta^2 x^{-4n-1}$$

NEET -2005

(Q). The displacement OF particle varies with time $x = a e^{-\alpha t} + b e^{\beta t}$ then find velocity

$x = a e^{-\alpha t} + b e^{\beta t}$

↓

$V = a(-\alpha) e^{-\alpha t} + b\beta e^{\beta t}$

$V = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$

with ↓ with time ↓

with ↑ with time ↑

JEE - 2023

Acceleration of object $a = 3t + 4$ where intial velocity is zero then velocity at $t = 2$ sec. done

NEET - 2025

(Q) Time is given as $t = x^2 + x$ then acceleration of the particle

$\longrightarrow$ $t = x^2 + x$

diff^n w.r.t x

$$\frac{dt}{dx} = \frac{dx^2}{dx} + \frac{dx}{dx}$$

$$\frac{1}{v} = (2x+1)$$

$$v = \frac{1}{(2x+1)} = (2x+1)^{-1}$$

$$a = v\frac{dv}{dx} = (2x+1)^{-1}\frac{d(2x+1)^{-1}}{dx}$$

$$= \frac{1}{(2x+1)} \times -1(2x+1)^{-2} \times 2$$

$$a = \frac{-2}{(2x+1)^3}$$

$$a = -2\left[\frac{1}{2x+1}\right]^3 = -2v^3$$

MQ*Q

(Q) velocity of object $V = a\sin(\omega t)\,\hat{i} + a\cos(\omega t)\,\hat{j}$ then find distance travelled by object in 2sec.

H/w

Constant acceleration

Sare JEE Advan copy

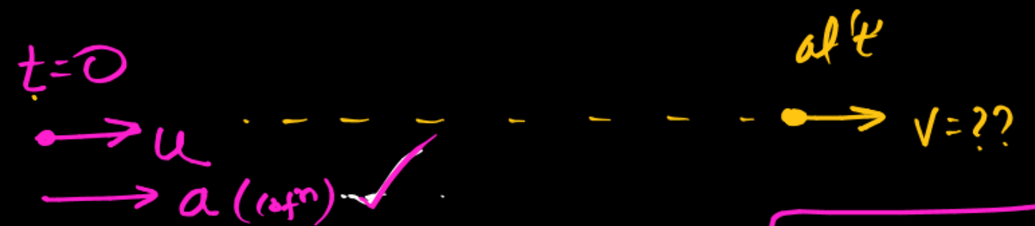
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rem:

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(1) Object starts his motion at $t=0$ with intial velocity u and constant acceleration a then find velocity at t & displacement in time t.



$$a = \frac{dv}{dt}$$

$$\int_u^v dv = \int_0^t a\,dt$$

$$[v]_u^v = a(t)_0^t = at$$

$$\boxed{\vec{v} = \vec{u} + \vec{a}t}$$

$$\boxed{\vec{v} = \vec{u} + \vec{a}t}$$

$$\frac{dx}{dt} = (u + at)$$

$$\int_{x_i}^{x_f} dx = \int_0^t u\,dt + \int_0^t at\,dt$$

$$(x)_{x_i}^{x_f} = ut + a\frac{t^2}{2}$$

$$\boxed{\vec{x}_f - \vec{x}_i = \vec{u}t + \frac{1}{2}at^2}$$

$$a = v\frac{dv}{dx}$$

$$\int_{x_i}^{x_f} a\,dx = \int_u^v v\,dv$$

$$\boxed{v^2 - u^2 = 2ax}$$

$\vec{s} = \vec{u}t + \frac{1}{2}at^2$

only valid for const^n acc^n

$\vec{V} = \vec{u} + \vec{a}t$

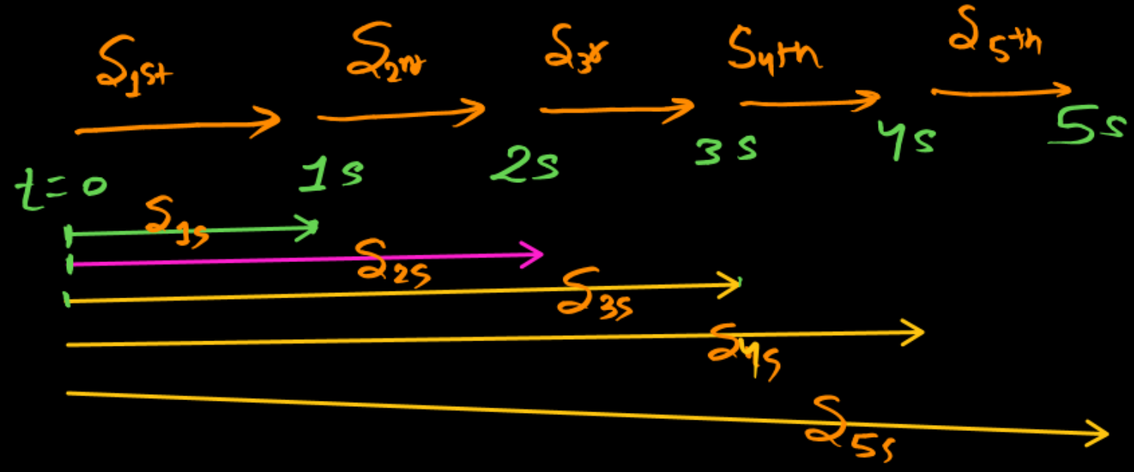
$s = \vec{u}t + \frac{1}{2}\vec{a}t^2$

disp^n in total time

$v^2 - u^2 = 2as$

$s_{n^{th}} = u + \frac{a}{2}(2n-1)$

$\vec{s} = \left(\frac{\vec{u}+\vec{v}}{2}\right)T$ ✓



(2) object moving with const^n acc $= 6 m/s^2$ & at $t=0$ $u = 2 m/s$ then find disp^m in 3sec & disp in 3rd sec.

$s_{t=3s} = ut + \frac{1}{2}at^2 = 2\times3 + \frac{1}{2}\times6\times(2)^2$

$= 6 + 3\times4 = 18m$

$s_{3rd} = u + \frac{a}{2}(2n-1)$

$= 2 + \frac{6}{2}(2\times3-1) = 2 + 3(5) = 17m$ ✓

(3) Object starts his mothion with initial velocity u after some time its velocity becomes v then find Average velocity. if accn is cutn $= a$



$v = u + at$ — ①

$v - u = at$

$$\text{Avg velocity} = \frac{\text{Total displ}^m}{\text{total time}}$$

$$\vec{V}_{Avg} = \frac{ut + \frac{1}{2}at^2}{t} = u + \frac{at}{2} = u + \frac{v-u}{2}$$

$$= \frac{2u+v-u}{2} = \frac{\vec{u}+\vec{v}}{2}$$

$\vec{s} = \left(\frac{\vec{u}+\vec{v}}{2}\right) T$

$\frac{\vec{s}}{T}$

$$\vec{V}_{Avg} = \frac{\vec{u}+\vec{v}}{2}$$

Question

Object starts his motion with velocity 10 m/sec after 8 sec its velocity becomes 24 m/sec then find total displacement in this interval.(a = constant)

$t=0$

$u = 10 m/s$ ✓

$t = 8 sec \longrightarrow V = 24 m/s$ ✓

$$S = ut + \frac{1}{2}at^2$$

$$S = 10\times 8 + \frac{1}{2}a(8)^2$$

$$= 80 + \frac{1}{2}\frac{7}{4}(8\times 8)$$

$$= 80 + 56$$

$$= 136 m$$

$$V = u + at$$

$$24 = 10 + a\times 8$$

$$14 = 8a$$

$$a = \frac{14}{8} = \frac{7}{4}$$

MP

$$S = \left(\frac{10+24}{2}\right)\times 8$$

$$= 34\times 4$$

$$= 136 m$$ ✓

Question

No copy

Object starts his motion with velocity 10 m/sec and uniform acceleration=2 ms^2 then find velocity and displacement in 5 sec.

$v = u + at$

$s =$

Question

$t=0$
$u=0$

Object starts his motion from rest and acceleration = 5 m/s^2 then find displacement in 4 sec and in 4th sec.

$$S = ut + \frac{1}{2}at^2$$

$$= 0 \times 0 + \frac{1}{2} 5 (4)^2$$

$$S_{4^{th}} = u + \frac{a}{2}(2n-1)$$

Question

Object starts his motion with velocity u and acceleration a then find displacement in t^{th} sec.

Question

Object starts his motion from rest, constant 'a' then find ratio of displacement in n^{th}-sec and n-sec. [IIT/NEET]

$S_{n\text{-sec}} = un + \frac{1}{2}an^2$

$S_{n^{th}} = u + \frac{a}{2}(2n-1)$

Q.1 object starts his motn from rest ($u=0$) & constant acceleration $a = 10 m/s^2$ then find

displacment in

$S_{1s} =$

$S_{2s} =$

$S_{3s} =$

$S_{4s} =$

$S_{5s} =$

displ^{n} in

$S_{1^{st}} =$

$S_{2^{nd}} =$

$S_{3^{rd}} =$

$S_{4^{th}} =$

$S_{5^{th}} =$

Question

Object starts his motion from rest and constant acceleration then find ratio of displacement in 6th sec and 6 sec.

Question

Object starts his motion from rest and constant acceleration 'a' then find displacement in 1 sec, 2 sec, 3 sec, 4 sec, so on...

Question

Object starts his motion from rest and constant acceleration 'a' then find displacement in 1st 2 sec (0 sec to 2 sec) and displacement in $t = 2$ sec to $t = 4$ sec.

Question

If object starts his motion from rest and constant acceleration then its displacement and distance will be same?

Question

Object starts his motion from rest and constant acceleration then find ratio of distance in 1-sec, in 2-sec and in 3-sec.

Question

If $u = 0$ (initial velocity) and constant acceleration then find ratio of distance in 4-sec and 8-sec.

Question

Object starts his motion from rest and constant acceleration then find displacement in 5 sec and in next 5 sec?

Question

Object starts from rest and constant acceleration it moves 80 m in 6-sec then find displacement in 12-sec.

Question

Object starts his motion from rest and constant acceleration then find ratio of displacement in 1st s : 2nd s : 3rd s.

Question

A particle experiences a constant acceleration for 20 sec after starting from rest. If it travels a distance S_1 in the first 10 sec and a distance S_2 in the next 10 sec, then:

1. $S_1 = S_2$
2. $S_1 = S_2/3$
3. $S_1 = S_2/2$
4. $S_1 = S_2/4$

Question

A body starts from rest. What is the ratio of the distance travelled by the body during the 4th and 3rd second.

Question

Object starts his motion from rest and constant acceleration and moves 100 m in 1st 5-sec then find displacement in next 5-sec.

Question

An object accelerates from rest to a velocity 27.5 m/s in 10 sec then find distance covered by object in next 10 sec.

1. 550 m
2. 137.5 m
3. 412.5 m
4. 275 m

Question

Object starts his motion from u and constant acceleration then find velocity at mid point if velocity at end point is V.

Question

Object starts his motion with u and constant acceleration a then find its velocity at one 3rd displacement of complete journey if final velocity is V.



Stopping Distance

Question

A motor car moving with a uniform speed of 20 m/sec comes to stop on the application of brakes after travelling a distance of 10 m. Its acceleration is:

1. $20\ m/sec^2$
2. $-20\ m/sec^2$
3. $-40\ m/sec^2$
4. $+2\ m/sec^2$

Question

A car moving with a speed of 50 km/h can be stopped by applying brakes after atleast 6 m. If the same car is moving with a speed of 100 km/h, what is the minimum stopping distance?

1. 6 m
2. 12 m
3. 18 m
4. 24 m

Question

What will be the ratio of the distance moved by a freely falling body from rest in 4th and 5th seconds of journey? **[1989]**

1. 4 : 5
2. 7 : 9
3. 16 : 25
4. 1 : 1

Question

A car is moving along a straight road with a uniform acceleration. It passes through two points P and Q separated by a distance with velocity 30 km/h and 40 km/h respectively. The velocity of the car midway between P and Q is **[1988]**

1. 33.3 km/h
2. $20\sqrt{2}$ km/h
3. $25\sqrt{2}$ km/h
4. 35 km/h.

Question

A particle start moving from rest and constant acceleration. It travels a distance x in first 2s and y in 4s then correction relation is

1. $y = 3x$
2. $y = 4x$
3. $y = x$
4. $x = 4y$

THANK YOU